Predicting User Actions from UI Sequences

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PROBLEM OVERVIEW

GOAL

Classify user actions (click, drag, zoom, etc.) from UI interactions.

INPUTS

3 consecutive UI frames per sample → 147 features.

CHALLENGE

Severe class imbalance: Many classes < 1% of data, leads to biased predictions

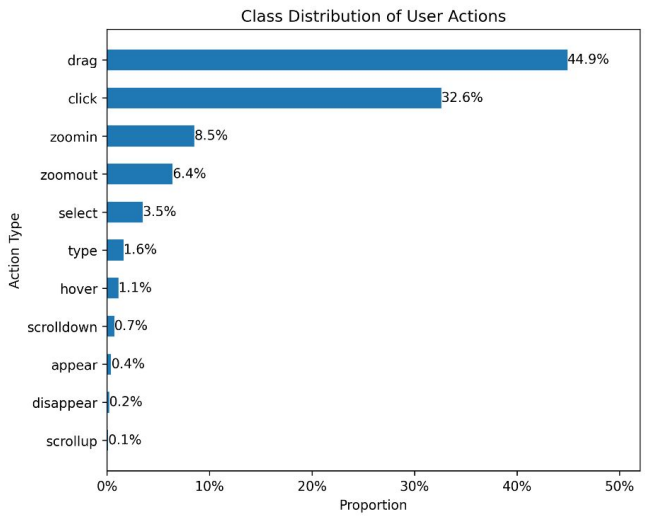
DATA OVERVIEW

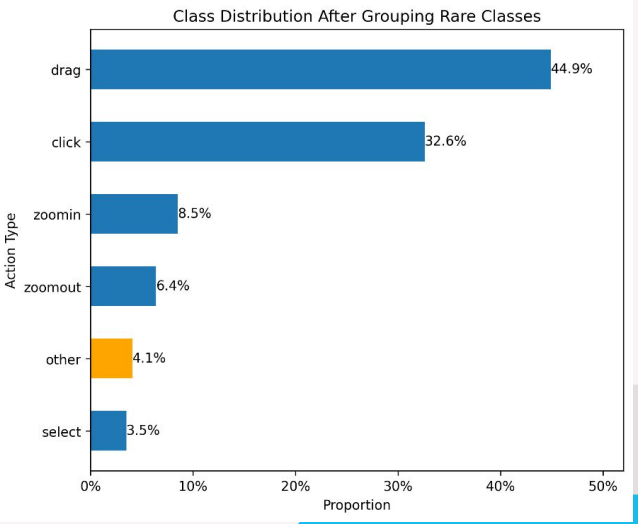
FOUR FILES

X_train, X_test, y_train, y_test.

CLASSES

11 original classes → grouped into 6 (with “other”).





APPROACHES + MODELS

01 One-Vs-All

- Built **6 logistic regression binary classifiers** (one per class).
- **Each model**: class vs all others.
- **Final prediction**: argmax of probabilities.
- Handles **multi-class + class-specific** boundaries.

02 Multi-Class Classification

- **Random Forest:** Robust to non-linear patterns and scaling differences.
- **XGBoost:** Boosting ensemble for improved imbalanced-class performance.

HANDLING IMBALANCE

MINORITY CLASS AGGREGATION

SMOTE

DIFFERENT CLASS WEIGHTS

MODEL + TRAINING

01 One-Vs-All

- **Scaled Features:** Standardized inputs.
- **Elastic Net Model:** Used logistic regression with L1 + L2 regularization.
- **Hyperparameter Tuning:** Optimized C and l1_ratio per class.
- **Threshold Tuning:** Class-specific thresholds on validation data to maximize precision/recall balance.

02 Random Forest

- **Hyperparameter Tuning:** Optimized tree depth, number of trees, split rules, and class weights.
- **Grid Search:** Retrieved the best model from GridSearchCV with 5-fold cross-validation.
- **Macro F1 Optimization:** Chose the best model based on macro-F1

03 XGBoost

- **Modeling:** Trained a single multi-class XGBoost model to predict all six action classes simultaneously.
- **Optimization & Evaluation:** Tuned hyperparameters using GridSearchCV with 5-fold cross-validation.

RESULTS: ONE-VS-ALL

Combined Multi-class Performance

Metric	Value
Accuracy	0.5596
Weighted Precision	0.6409
Weighted Recall	0.5596
Weighted F1	0.5849

Per-Class Performance Metrics

Class	Precision	Recall	F1
click	0.55	0.49	0.52
drag	0.75	0.54	0.63
other	0.14	0.5	0.22
select	0.12	0.48	0.19
zoomin	0.61	0.84	0.71
zoomout	0.7	0.73	0.72

STRONG MAJOR CLASSES

High performance on frequent classes (drag, zoomin, zoomout), balanced precision and recall.

WEAK MINORITY CLASSES

Low precision for rare classes (other, select), indicating many false positives.

MODERATE OVERALL PERFORMANCE

F1 score of 0.58 shows reasonable performance but inconsistency across classes.

RESULTS: RANDOM FOREST

STRONG MAJOR CLASSES

Zoom actions (zoomin, zoomout) achieved highest F1-scores (**0.84–0.85**).

WEAK MINORITY CLASSES

Minority classes (select, other) showed lower precision due to limited samples.

BETTER OVERALL PERFORMANCE

Accuracy	0.6856
Weighted Precision	0.7315
Weighted Recall	0.6856
Weighted F1-score	0.6936

	Precision	Recall	F1-score
click	0.59	0.72	0.65
drag	0.84	0.60	0.70
other	0.47	0.80	0.59
select	0.25	0.61	0.36
zoomin	0.78	0.91	0.84
zoomout	0.86	0.84	0.85

RESULTS: XGBoost

Class	Precision	Recall	F1-Score
click	0.65	0.71	0.68
drag	0.85	0.64	0.73
other	0.25	0.65	0.37
select	0.25	0.57	0.34
zoomin	0.84	0.90	0.87
zoomout	0.70	0.84	0.76

Accuracy	Precision (weighted)	Recall (weighted)	F1 (weighted)
0.6934	**0.7422**	**0.6934**	**0.7062**

Strong Performers

The zoom-in class performs best with an **F1 of 0.87** (precision 0.84, recall 0.90), followed by zoom-out with an **F1 of 0.76.**

Moderate Performers

Drag achieves an **F1 of 0.73** with very high precision (0.85) but lower recall (0.64), while click shows a more balanced performance with an **F1 of 0.68.**

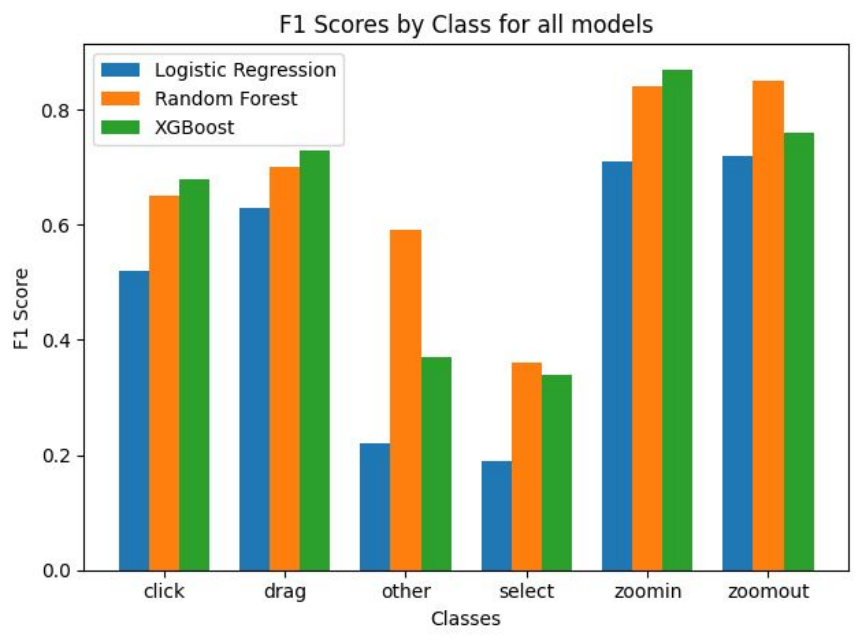
Weak Performers

The "other" (F1 0.37) and "select" (F1 0.34) classes perform worst due to low precision and limited sample sizes, leading the model to over-predict these categories.

KEY TAKEAWAYS

IMBALANCE DRIVES PERFORMANCE

Model works well on major classes but struggles on rare classes due to class imbalance.

TREE-BASED MODELS PERFORM BETTER

XGBoost and Random Forest handle complex patterns and feature interactions better.